PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GORVEMENT
FORM IV TAHOSSA NORTHWESTERN ZONE MOCK EXAMINATION 2025

032/2A **CHEMISTRY 2A (ACTUAL PRACTICAL A)**

TIME 2:30HRS **AUGUST 2025**

INSTRUCTIONS

1. This paper consists of two (2) questions. Answer all questions
2. Each question carries twenty-five (25) marks
3. Qualitative Analysis Guide sheet authorized by NECTA and non – programmable calculators may be used.
4. Cellular phones are not allowed in the examination room
5. Write your examination number on every page of your answer booklet(s)
6. Your may use the following constant:

Atomic masses:
C = 12, H = 1, Ag = 108, Na = 23, Cl =35.5.
1litre = $1dm^3$ = $1000cm^3$.

1. You are provided with the following solutions:

MM: A solution of 0.M Hydrochloric acid
PP: A solution of impure sodium carbonate anhydrous made by dissolving 1.3g in $250cm^3$ of water
MO: Methyl Orange indicator.

Questions:

a) (i) Titrate solution MM against (20 or 25) cm^3 of PP until the colour change. Record the burette readings. Repeat the procedures to obtain three accurate readings and record your results in tabular form.

(ii) Why the colour of the solution in a conical flask changed at the end?

(iii)Determine the average litre volume.

(iv) .....cm^3 of solution MM required .....cm^3 of solution PP for complete neutralization.

(v) Assume that sulphuric acid of the same molarity was used in the place of hydrochloric acid , would it be different in the litre volume used? Give a reason

(b)(i) Name the apparatus you used for measuring volume of MM

(ii) Why the apparatus used in (b)(i) is best recommended for its function?

(c) Write balanced chemical equation for the reaction between MM and PP

(d) Calculate the following and write your answer into two decimal place.

(i) Molarity of PP
(ii) Percentage purity of PP

2. Sample N is a simple salt containing one cation and one anion. Carefully carry out all experiments described in the table below and record all your observations and make appropriate interences to identify the ions present in sample N.

S/N	Experiment	Observation	Inference
i)	Observe sample N		
ii)	Put a spatulaful of sample N in a test tube and add distilled water		
iii)	Transfer small portion of sample N in a test tube and add concentrated sulphuric acid.		
iv)	Transfer a Spatulaful of sample N in a test tube, add dilute hydrochloric acid		
v)	Use the resulting solution of sample N in (ii) above divide it into four (4) portion (a) To the first portion add NaOH solution drop wise till in excess		
	(b) To the second portion add ammonia solution dropwise till in excess		
	(c) To the third portion add potassium ferrocyanide solution		
	(d) To the fourth portion add $MgSO_4$		

Conclusion

(a) (i) The cation in a sample N is .....

(ii)The anion in sample N is .....

(iii)The compound N is .....

(b) Write the reaction equation that took place at experiment (iii) above

(c) State three uses of the gas produced at experiment (iii) above

(d) Identify any two significances of qualitative analysis in your daily life